



Technical Service Bulletin

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Glycerin Coolant Introduction

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin introduces the availability of glycerin coolant for Cummins® engines. Glycerin has been determined to be an acceptable base for antifreeze/coolant, and is now available from Cummins Filtration™. The product is identified as ES Compleat™ Glycerin Premix. The following table lists the Cummins Filtration™ part numbers for this product.

Part Number	Part Description	Container Volume
CC36004	ES Compleat™ Glycerin Premix BULK	N/A
CC36003	ES Compleat™ Glycerin Premix TOTE	1040 liter [275 gal]
CC36002	ES Compleat™ Glycerin Premix DRUM	208 liter [55 gal]
CC36001	ES Compleat™ Glycerin Premix PAIL	19 liter [5 gal]
CC36000	ES Compleat™ Glycerin Premix GALLON	3.785 liter [1 gal]

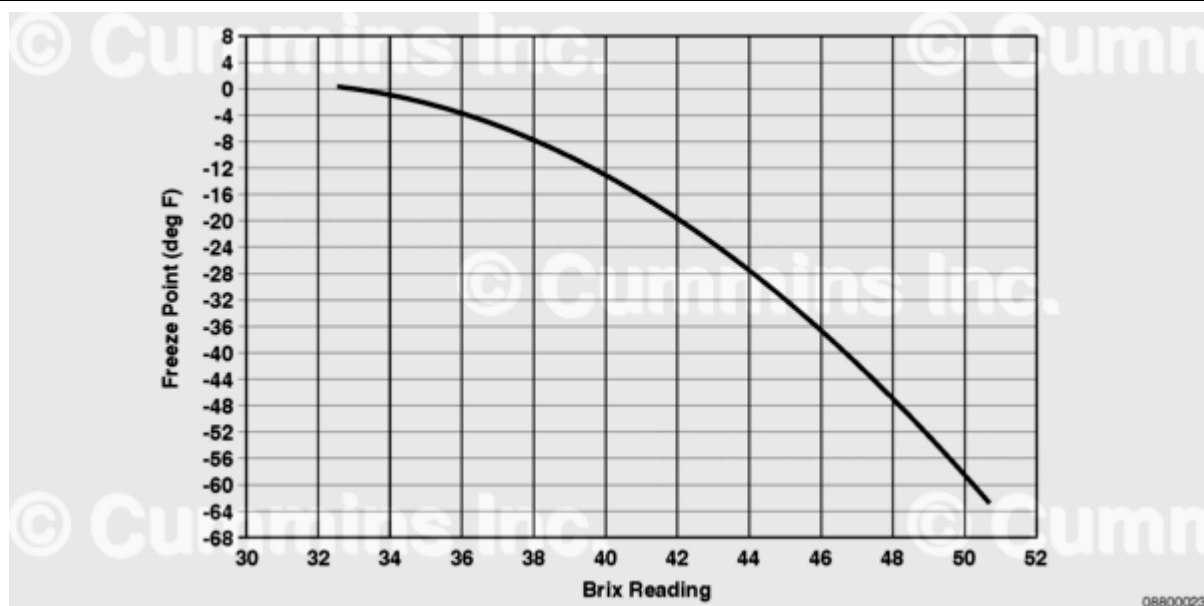
Glycerin is a by-product of biodiesel production. Cummins Inc. started evaluating it as an antifreeze base several years ago and has recently determined it to be an acceptable base for antifreeze/coolants used in Cummins® engines. Reference SAE paper 2007-01-4000 (An Evaluation of Glycerin (Glycerol) as a Heavy Duty Engine Antifreeze/Coolant Base) for more information on the subject. The most recent version of Application Engineering Bulletin (AEB) 99.01, titled Engine Coolant Performance Specifications, allows the use of glycerin as an antifreeze/coolant base. It **must** be stressed that additive packages used with glycol based

products, that meet AEB 99.01, can **not** be assumed to meet this standard when used with glycerin based products. See AEB 99.01 for more information. Similar to propylene glycol, glycerin is non-toxic and is also used in many food, medical, and cosmetic products.

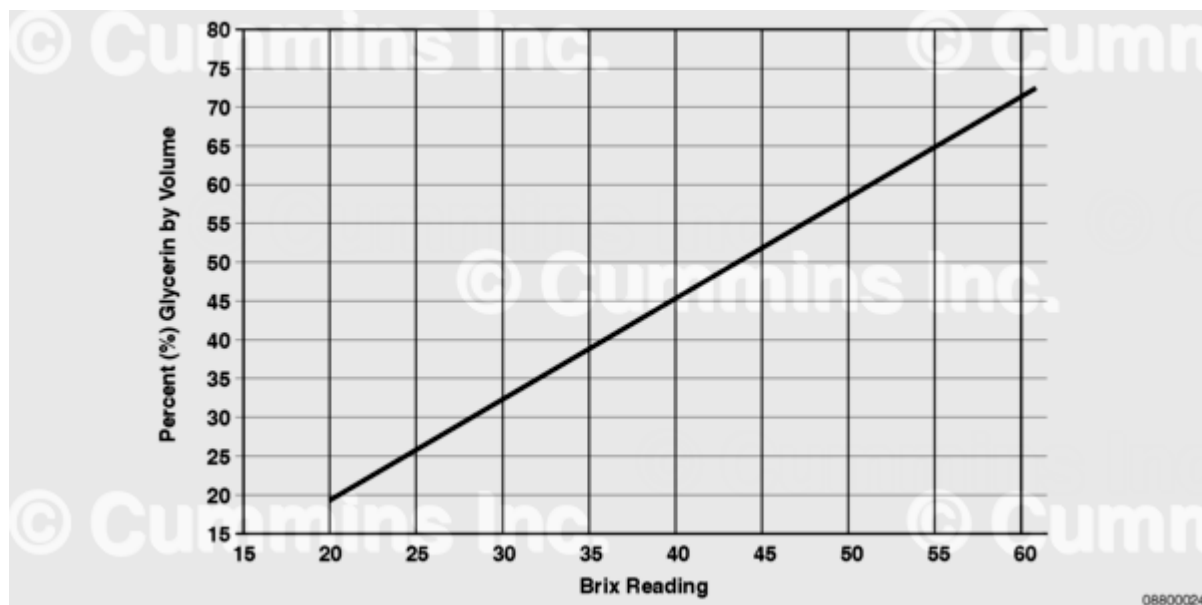
It is important to read and understand the following points when deciding to use or switch to glycerin based coolants:

- Glycerin based coolants **must** meet the most recent requirements of AEB 99.01.
- Due to its high viscosity at low temperatures, when mixed with water at the required ratios, glycerin based coolants are **not** recommended for use in arctic climates. Refer to Cummins® Coolant Requirements and Maintenance, Bulletin 3666132 (</qs3/pubsys2/xml/en/bulletin/3666132.html>) for details concerning what constitutes arctic conditions.
- No cooling system hardware or control changes are required when converting from glycol based coolants to glycerin based coolants. It is recommended to completely drain any glycol based coolant from the system before converting to glycerin based coolant.
- Glycerin based coolants are mixed with water at the same ratio (generally a 50/50 mixture) as glycol based coolants.

Test strips and refractometers presently used for evaluation of the conditions of glycol based coolants in service are **not** usable for glycerin based coolants. For evaluation of additive levels (nitrite and molybdate) for glycerin based coolant, use Cummins Filtration™ 2-way test strips, CC36050 (bottle of 50) or CC36050B (Single Foil). For evaluation of freeze point and glycerin concentration, use a Brix scale refractometer with an extended range, similar to Cummins Filtration™, Part Number CC36049. Use the following charts to convert the Brix reading to freeze points and glycerin concentrations.



Glycerin Coolant Brix Reading vs Freeze Point



Glycerin Coolant Brix Reading vs Percent Glycerin by Volume

Since there are presently no devices suitable for field use to measure and evaluate the condition of glycerin and glycol mixes, the use of glycerin coolants **must** be restricted to users who can completely control the coolant use in their equipment. It is important to note, that mixing glycerin and glycol based antifreeze/coolants does **NOT** cause any incompatibility problems, but makes it very difficult to assess the freeze point and additive levels of the mixture. If mixing of glycol and glycerin based coolants can **not** be avoided, coolant samples **must** be evaluated to determine freeze point and additive levels, using a commercial lab, that is able to measure these mixed coolants. Polaris Laboratories, LLC is set up to evaluate these mixtures. They can be contacted at www.polarislabs.com. Look for Fleetguard® Monitor C, CC2700 or CC2717 coolant analysis.